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(54) **SEMICONDUCTOR STRUCTURE AND MANUFACTURING METHOD THEREOF**

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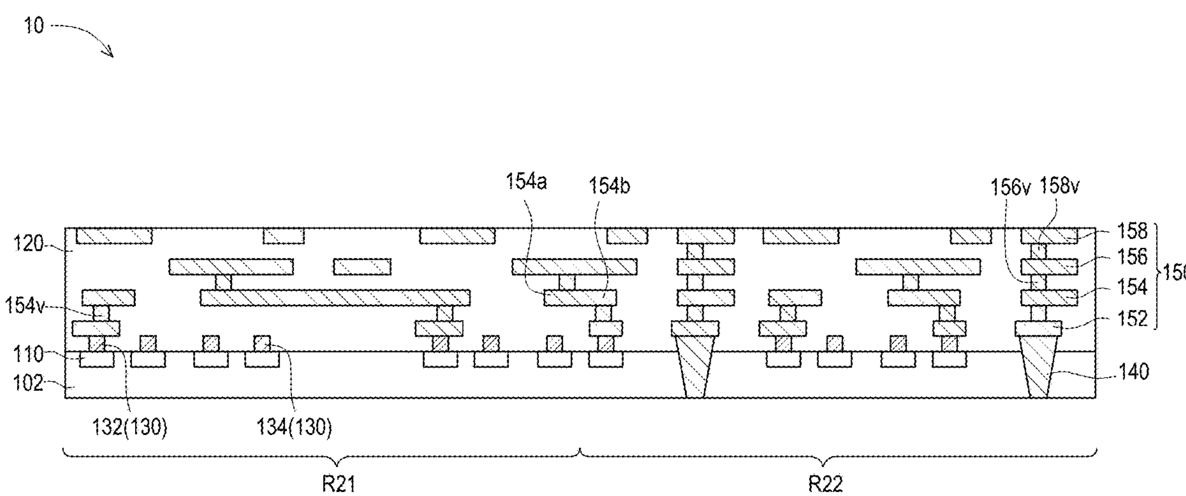
(57) **ABSTRACT**

Disclosed is a semiconductor structure and a manufacturing method thereof. The manufacturing method of the semiconductor structure includes the following steps. A plurality of passive elements are formed in a substrate. The plurality of passive elements are arranged in an array in the substrate. A plurality of contact members are formed on the plurality of passive elements. The plurality of contact members correspond to the plurality of passive elements. A redistribution layer is formed on the substrate. The redistribution layer is electrically connected to a part of the plurality of passive elements through a part of the plurality of contact members.



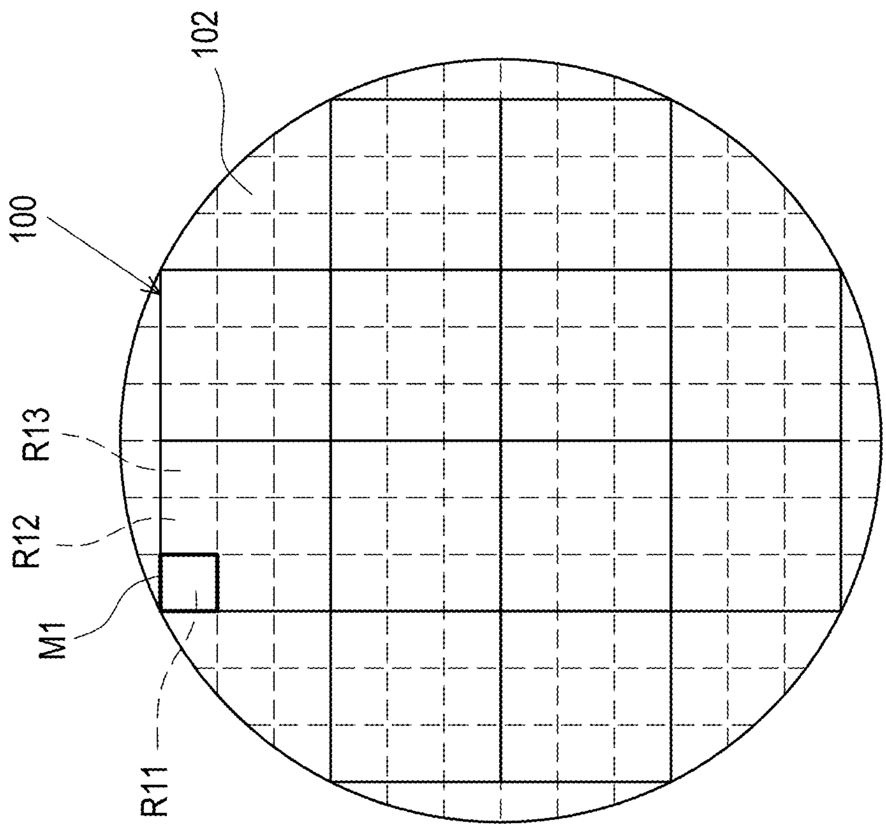


FIG. 1A

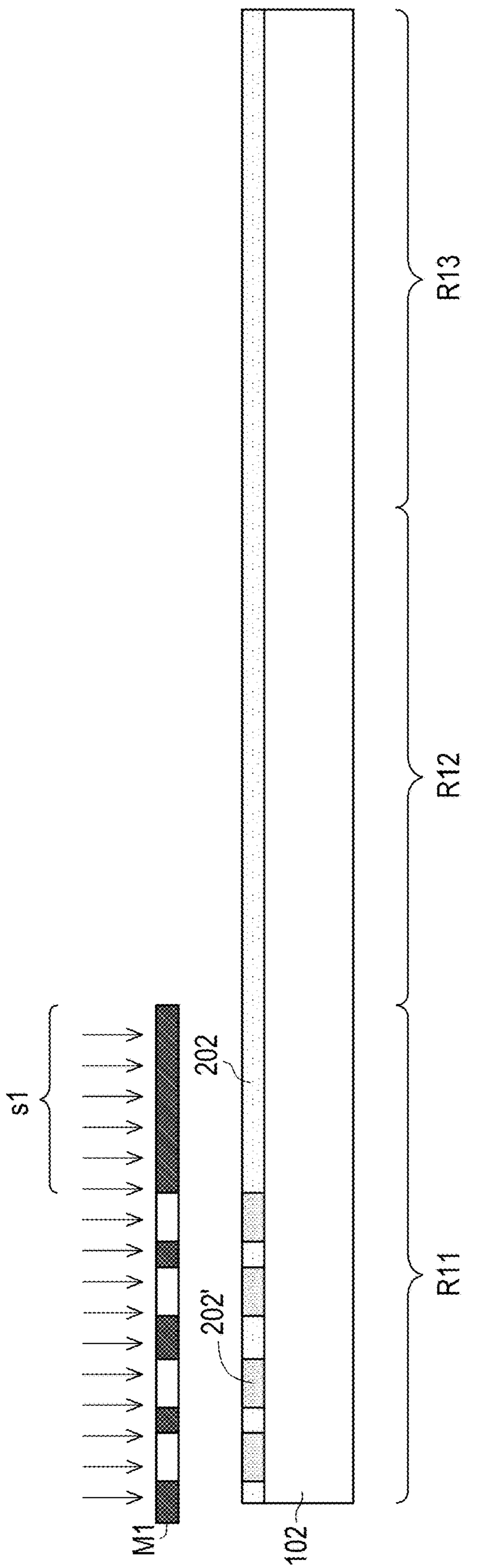


FIG. 1B

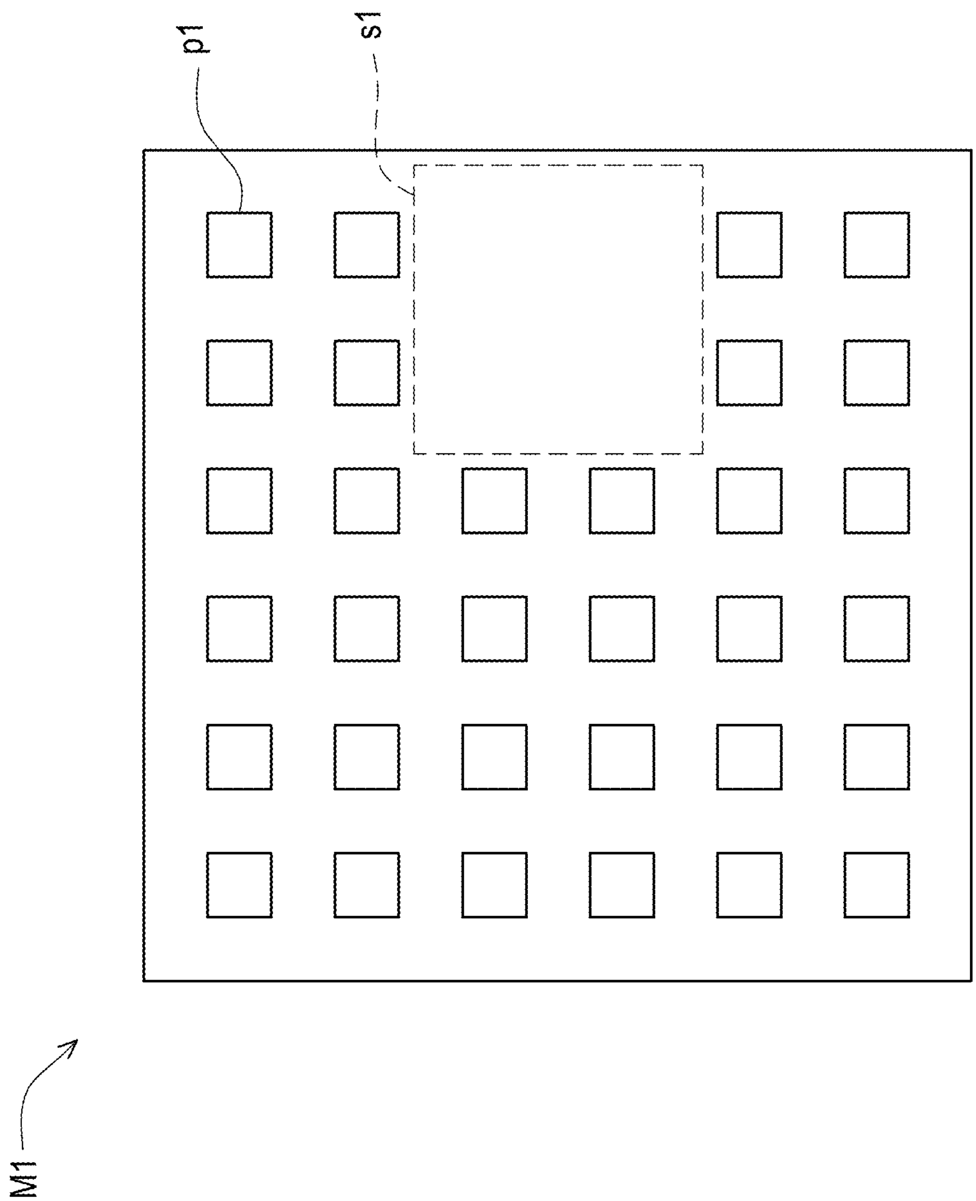


FIG. 1C

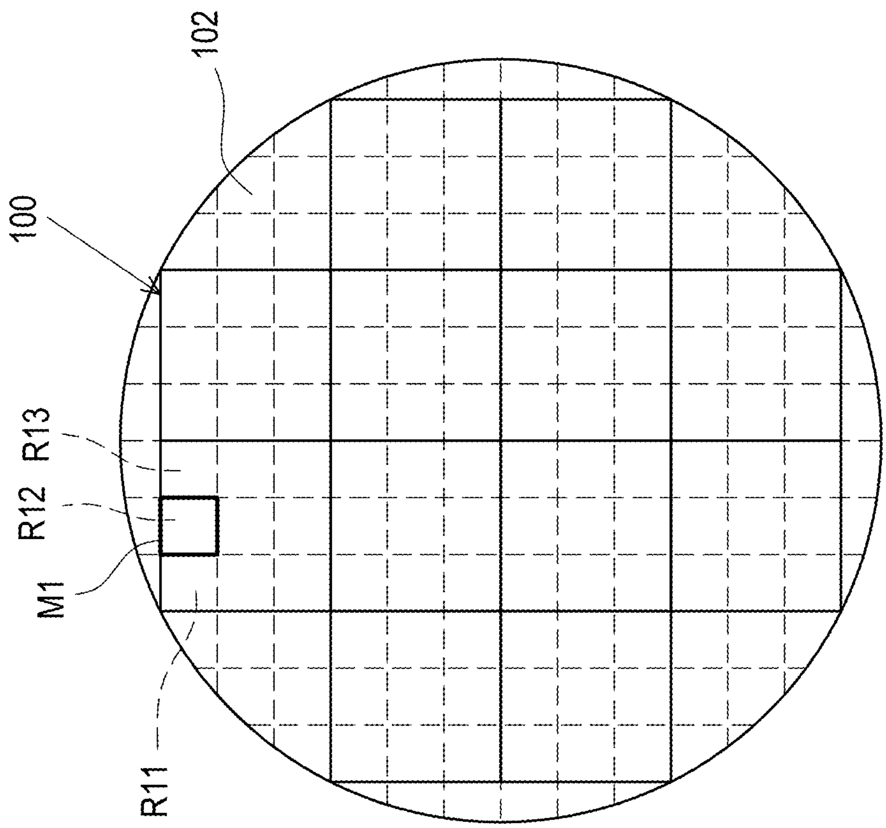


FIG. 2A

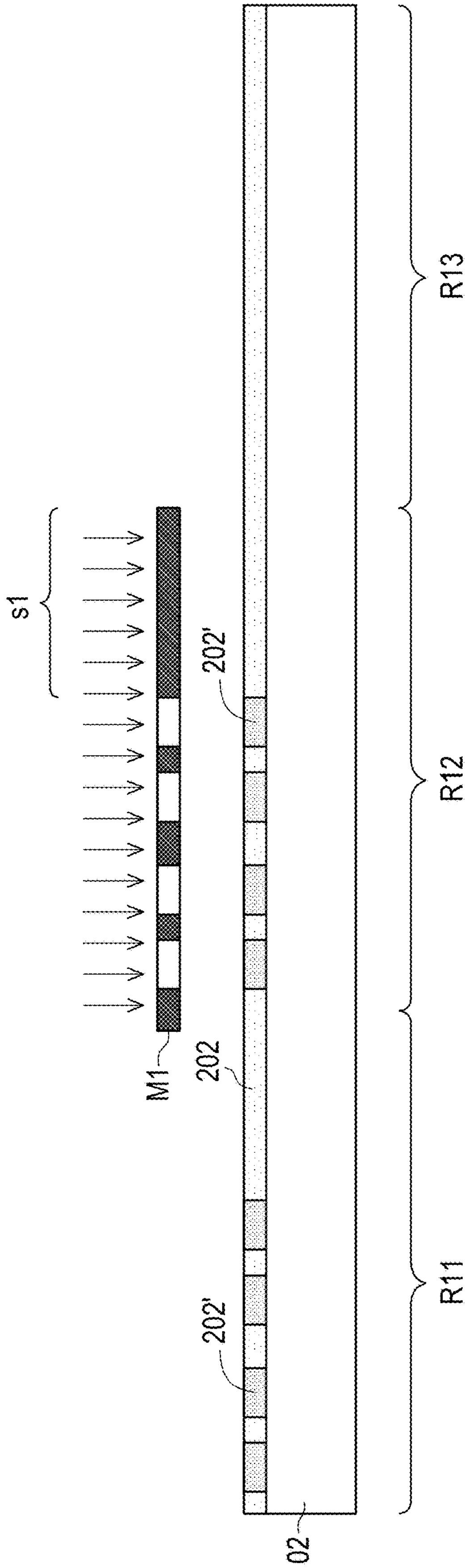


FIG. 2B

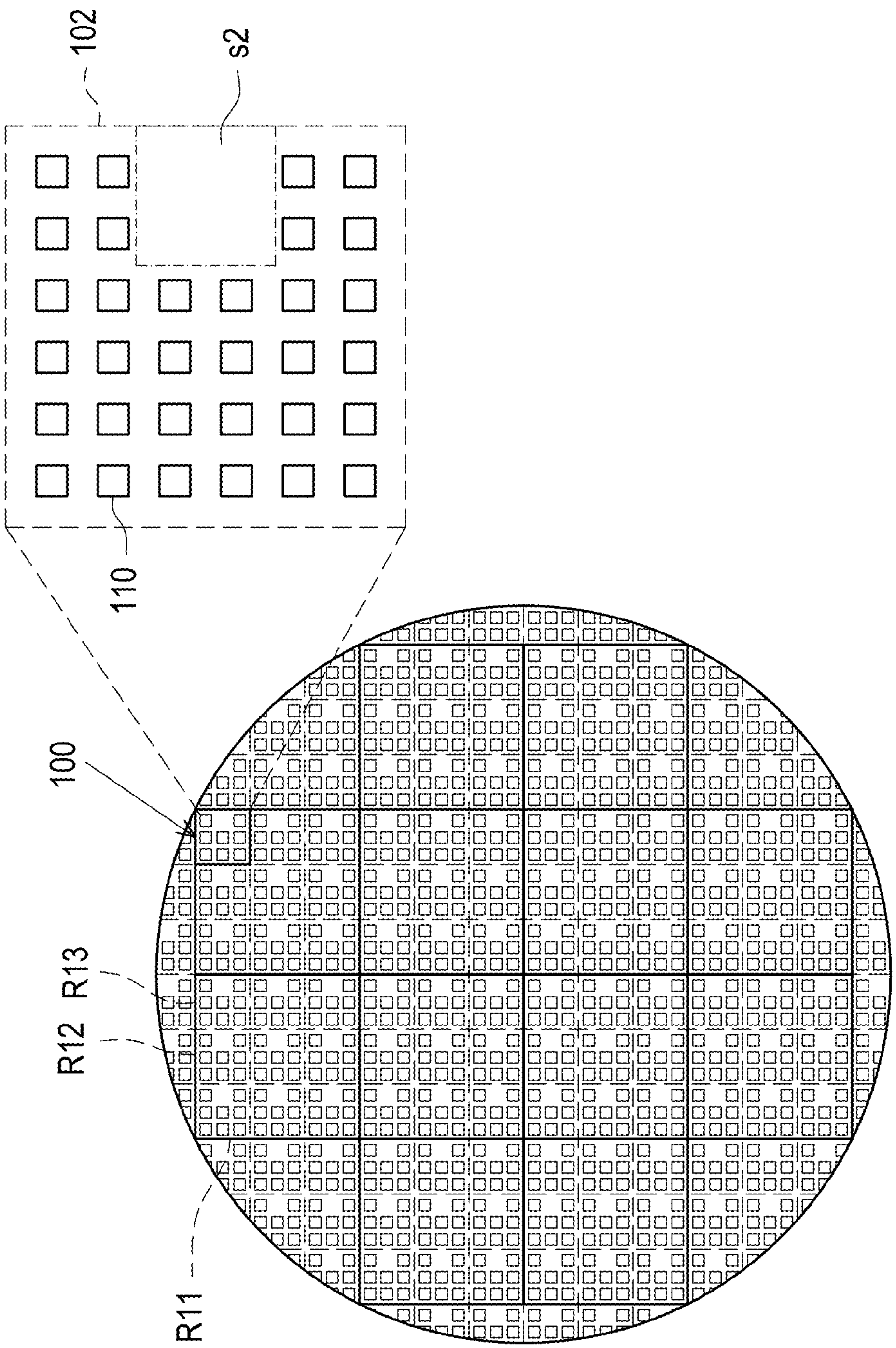


FIG. 3A

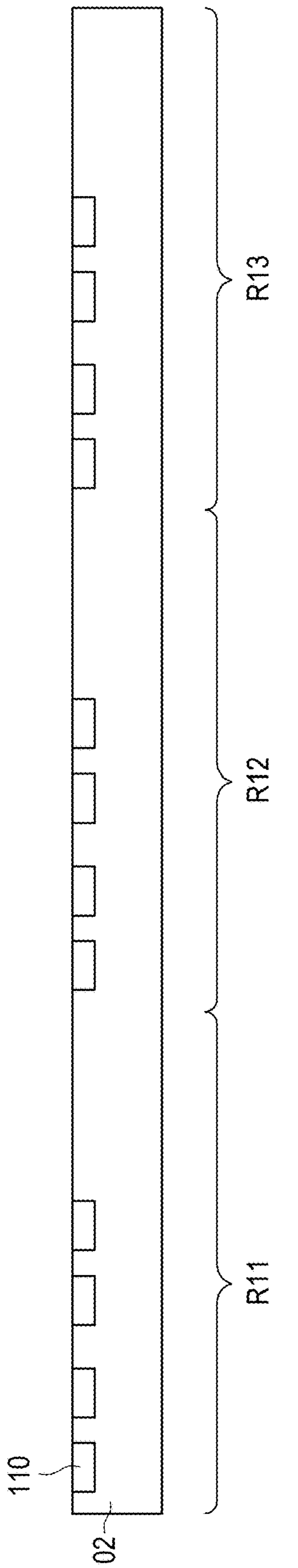


FIG. 3B

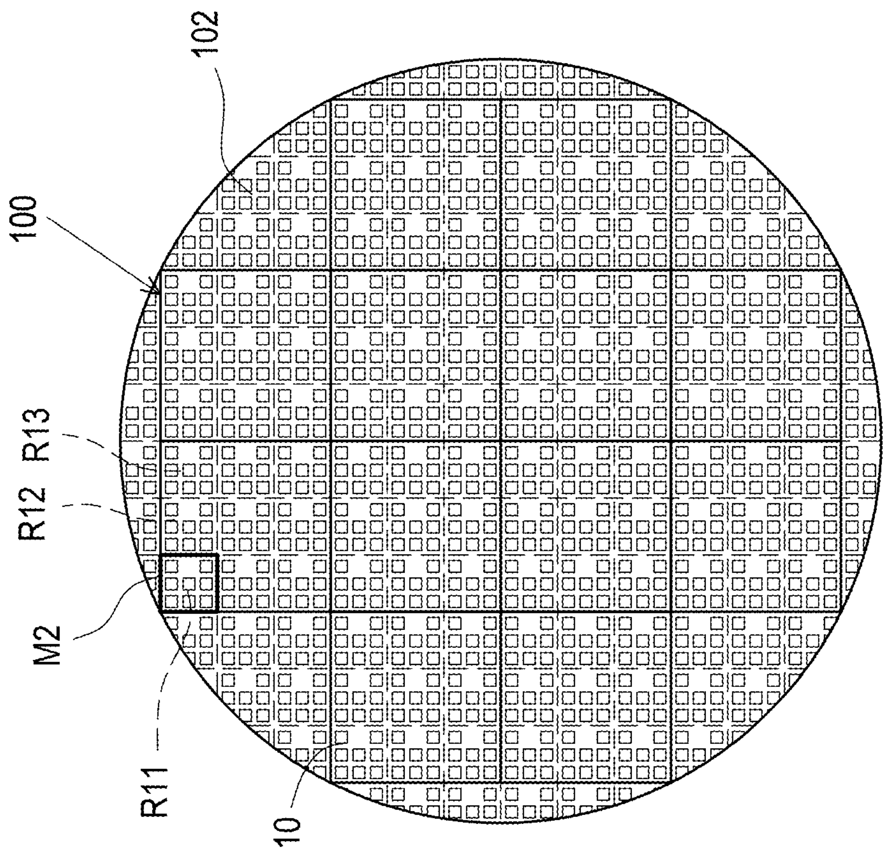


FIG. 4A

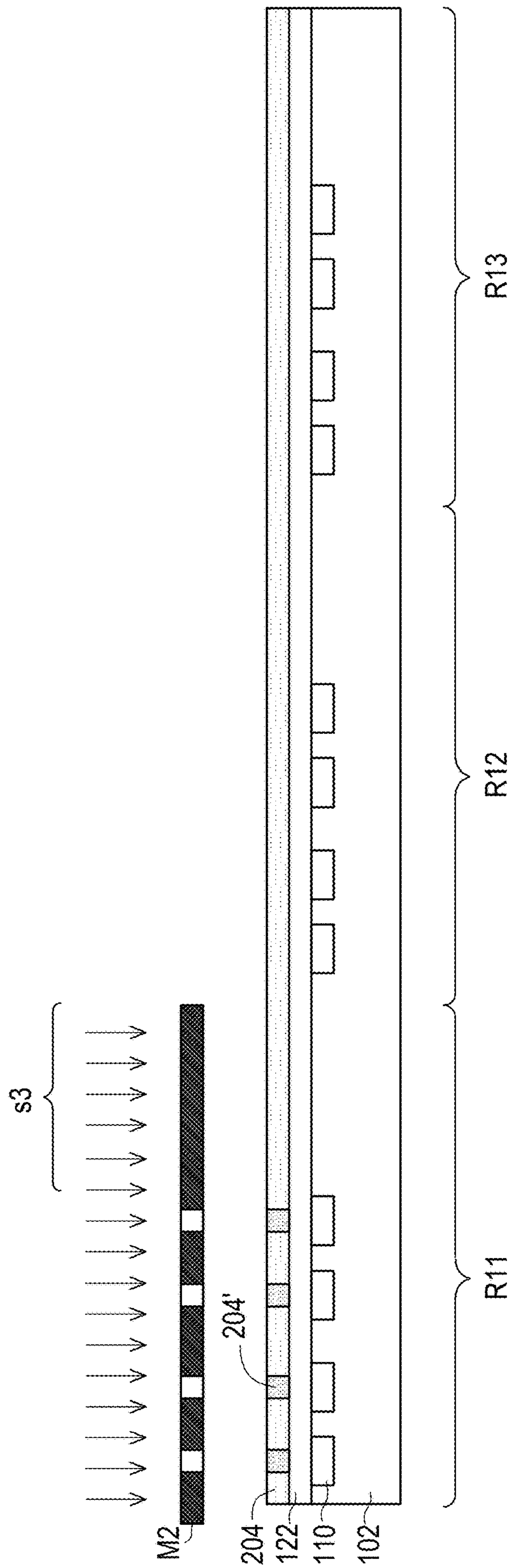


FIG. 4B

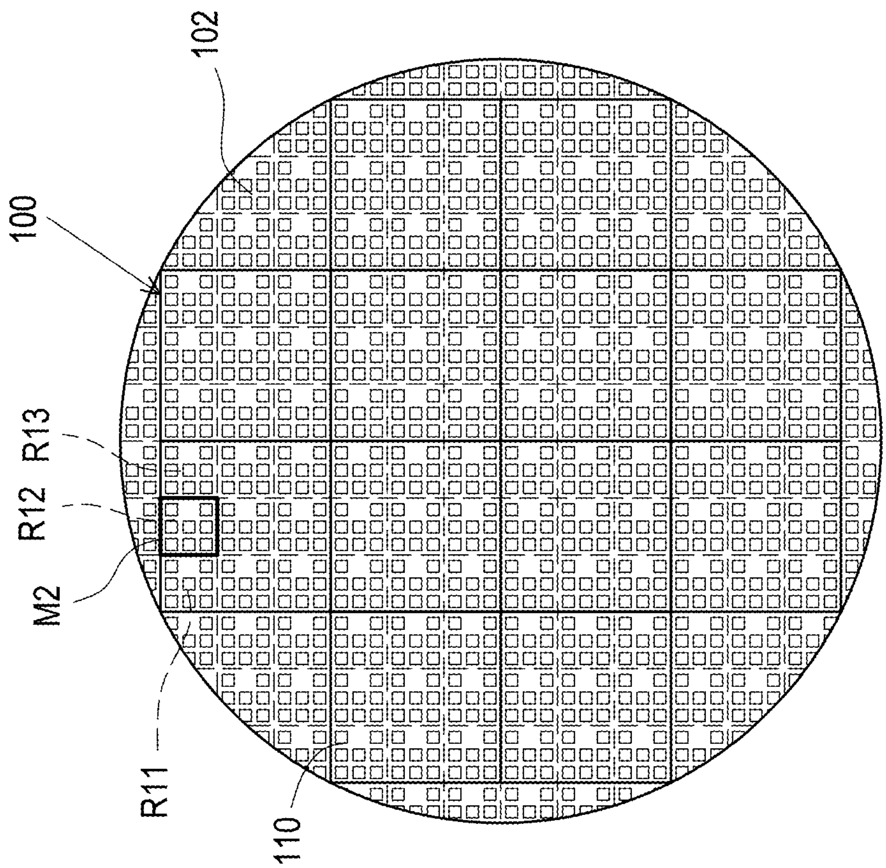


FIG. 5A

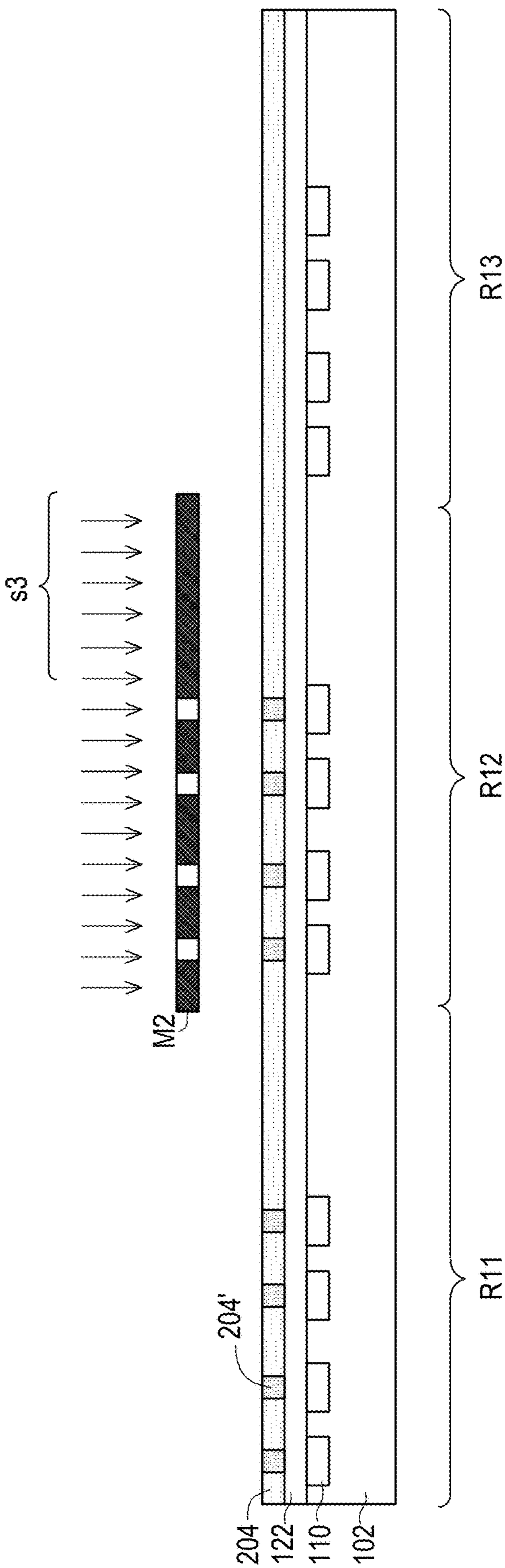


FIG. 5B

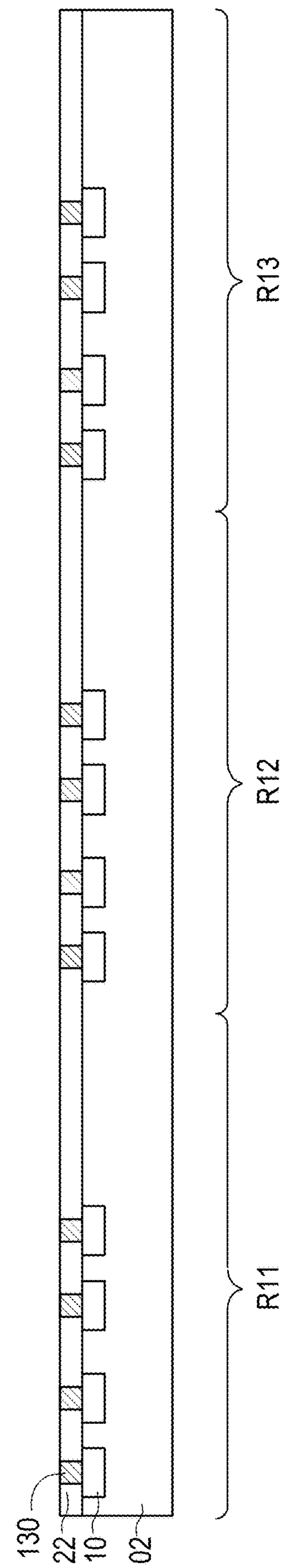


FIG. 6

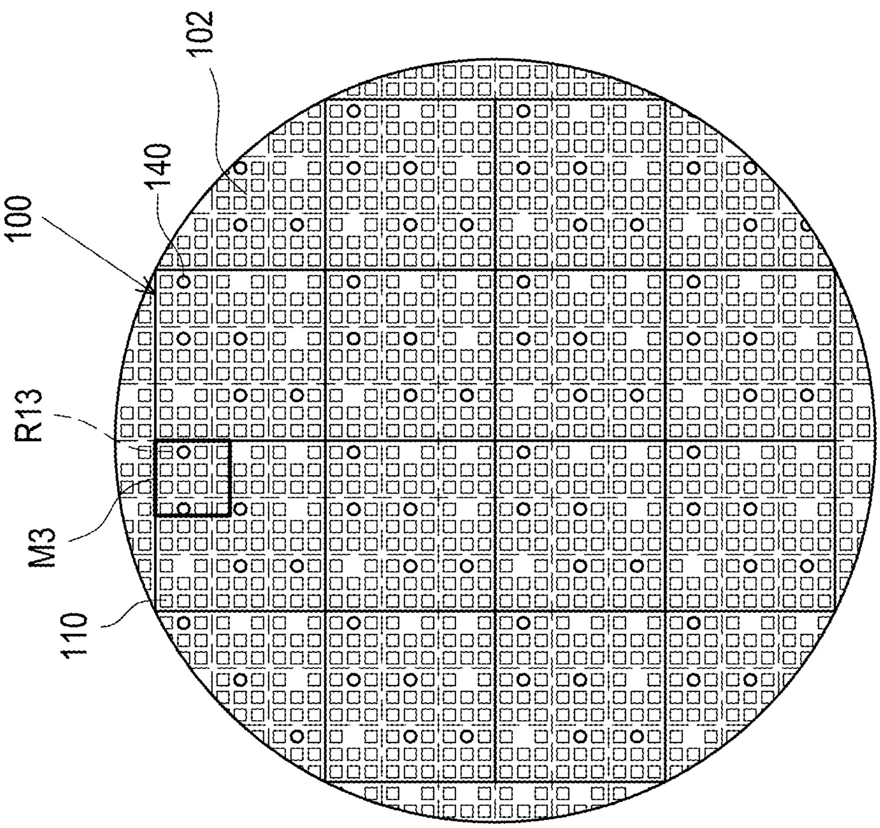


FIG. 7A

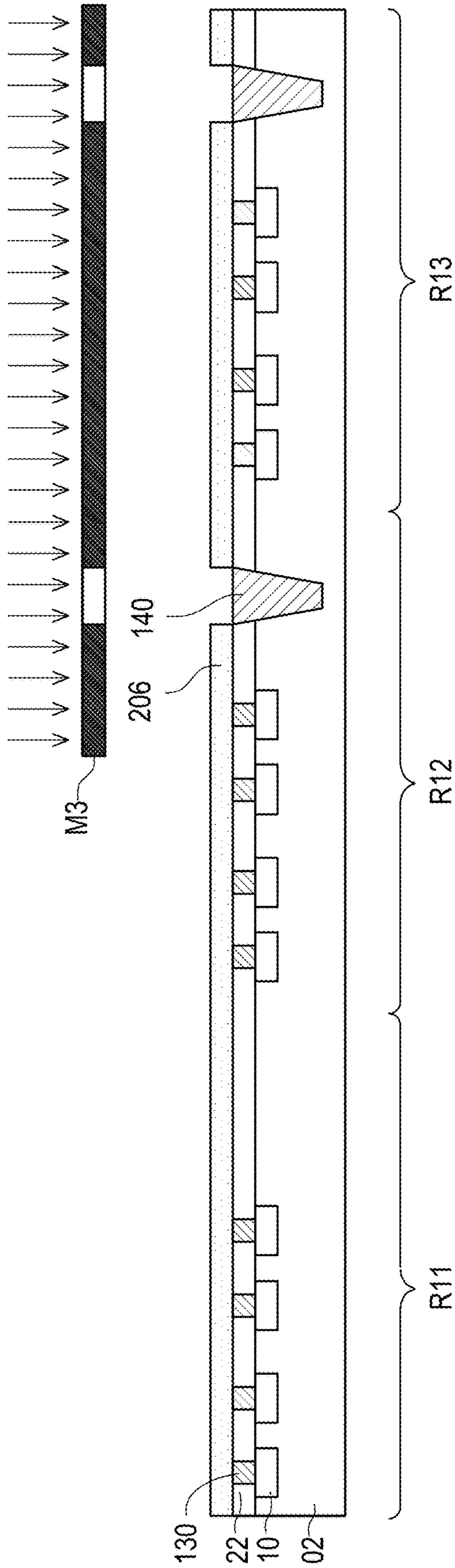


FIG. 7B

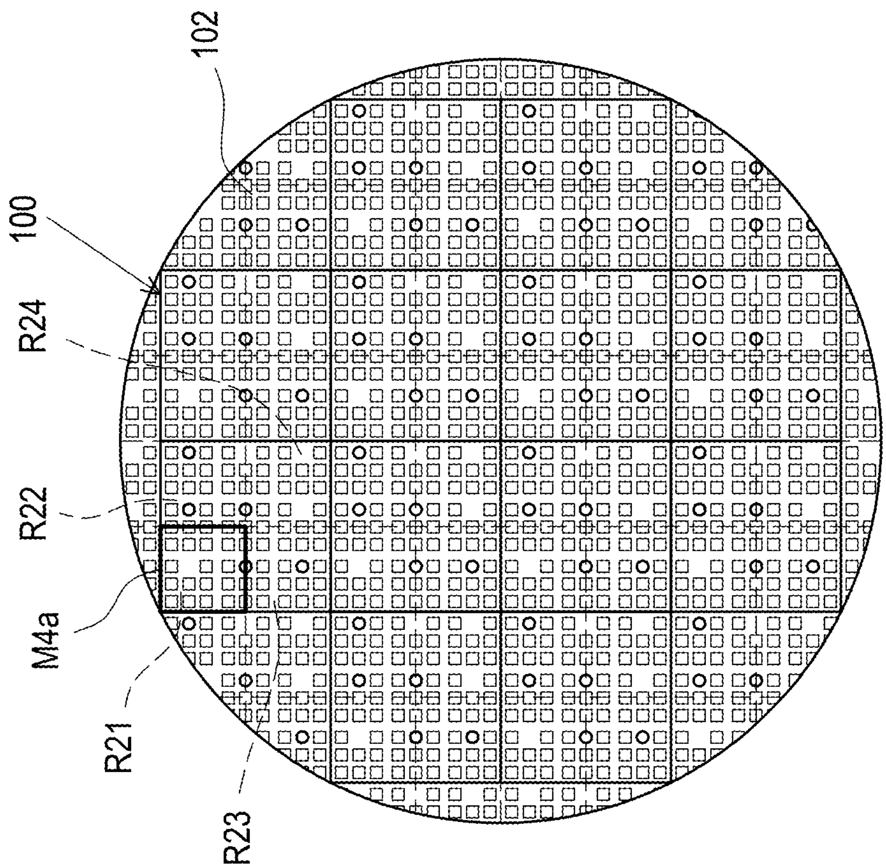


FIG. 8A

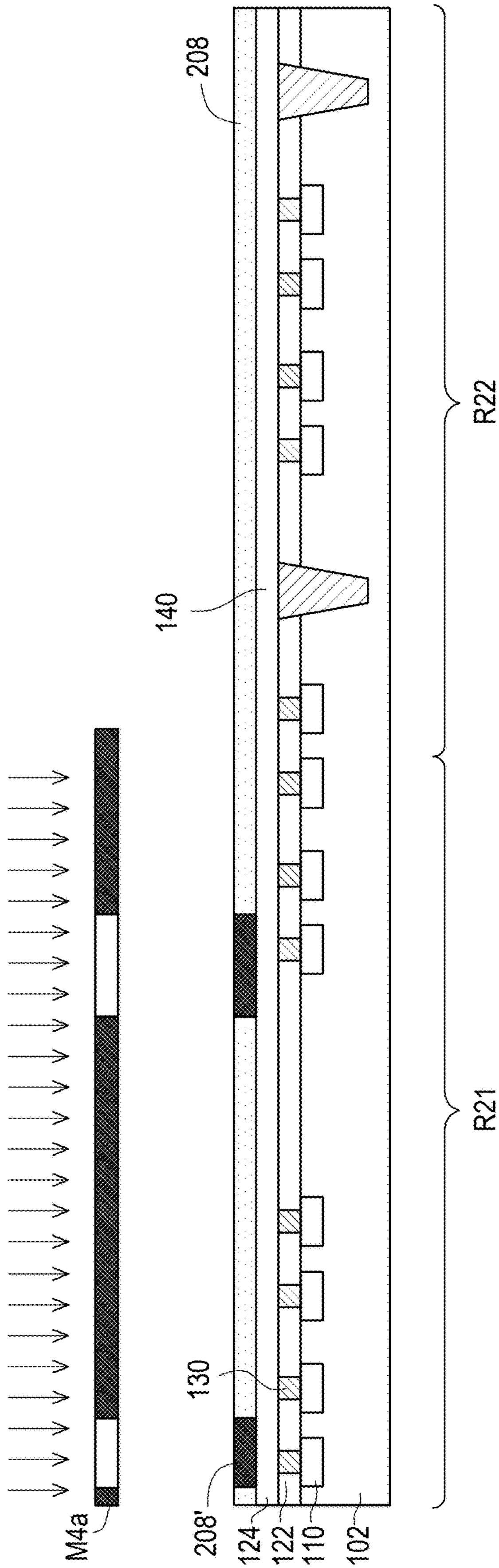


FIG. 8B

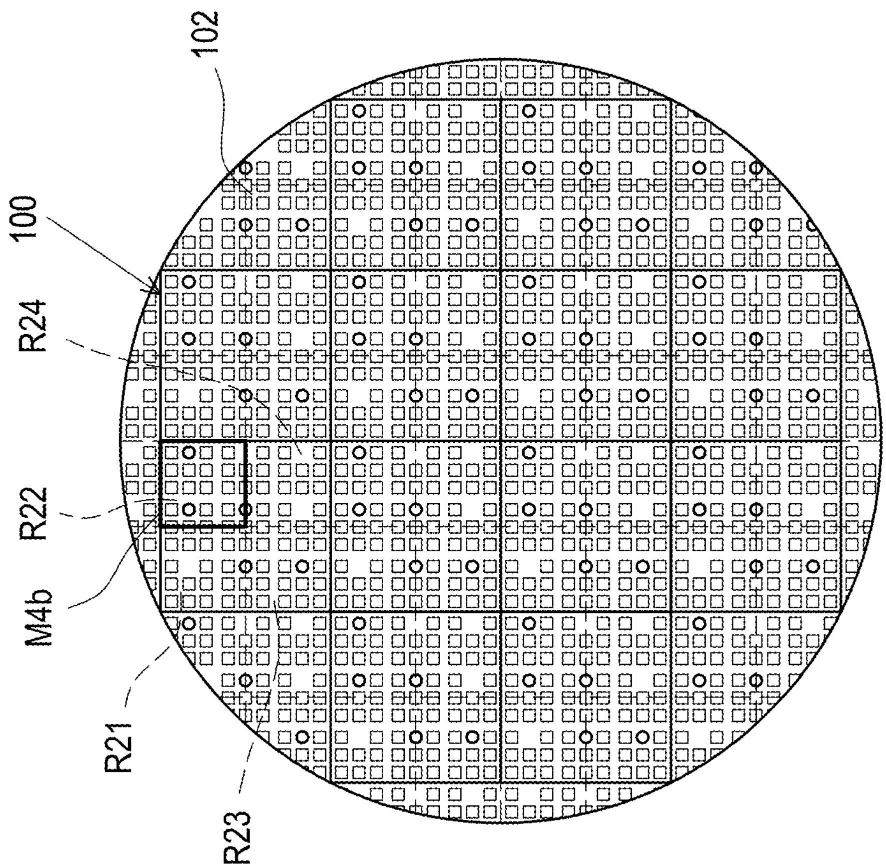


FIG. 9A

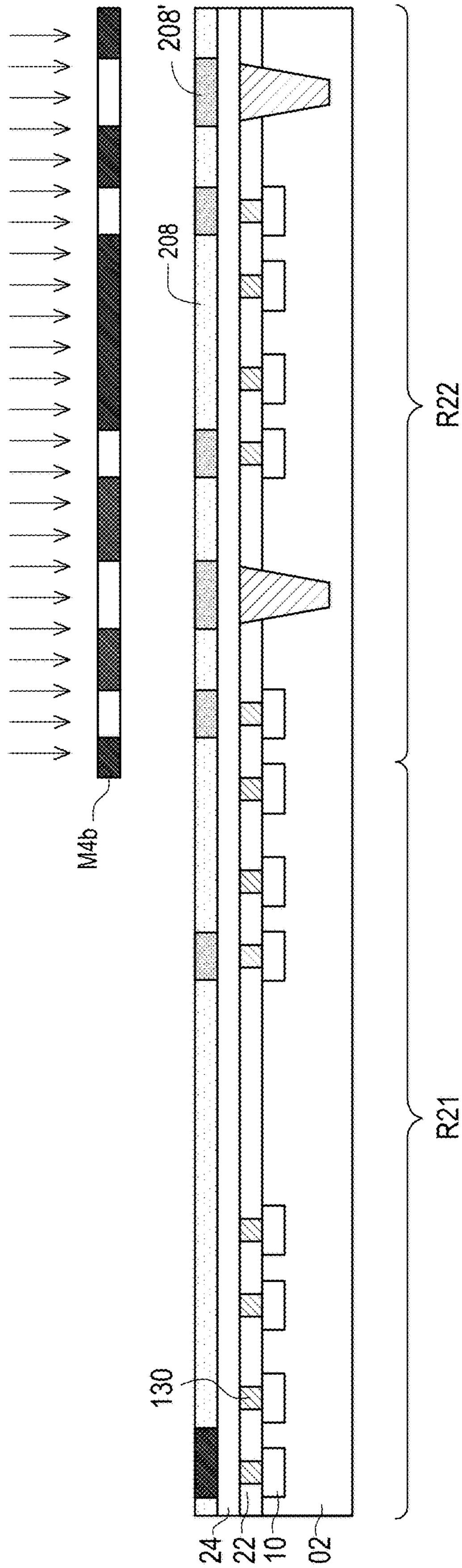


FIG. 9B

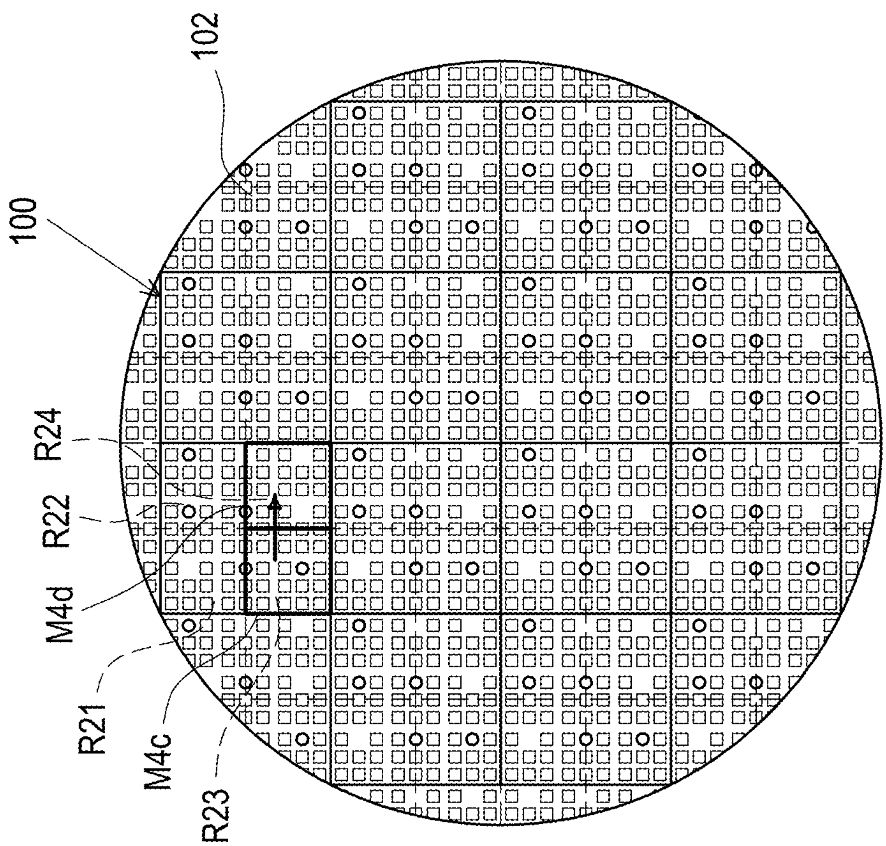


FIG. 10

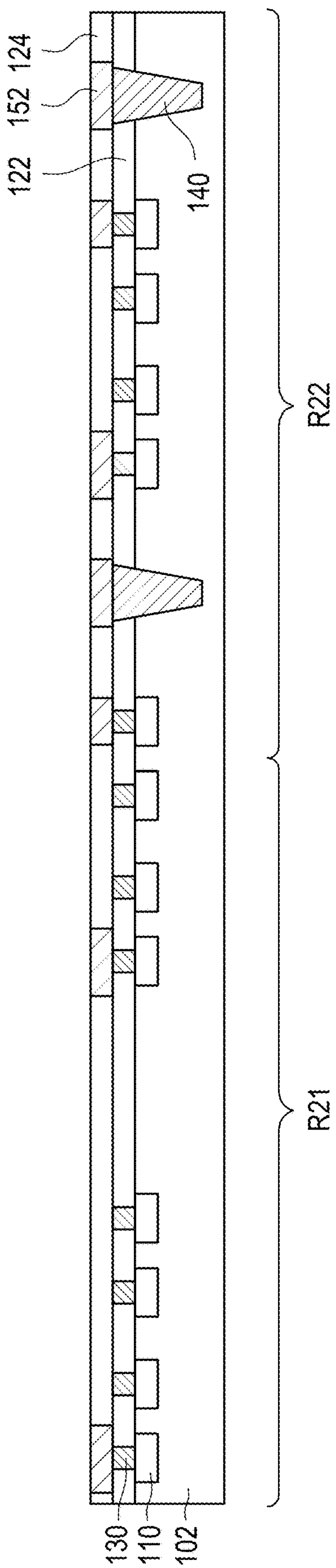


FIG. 11

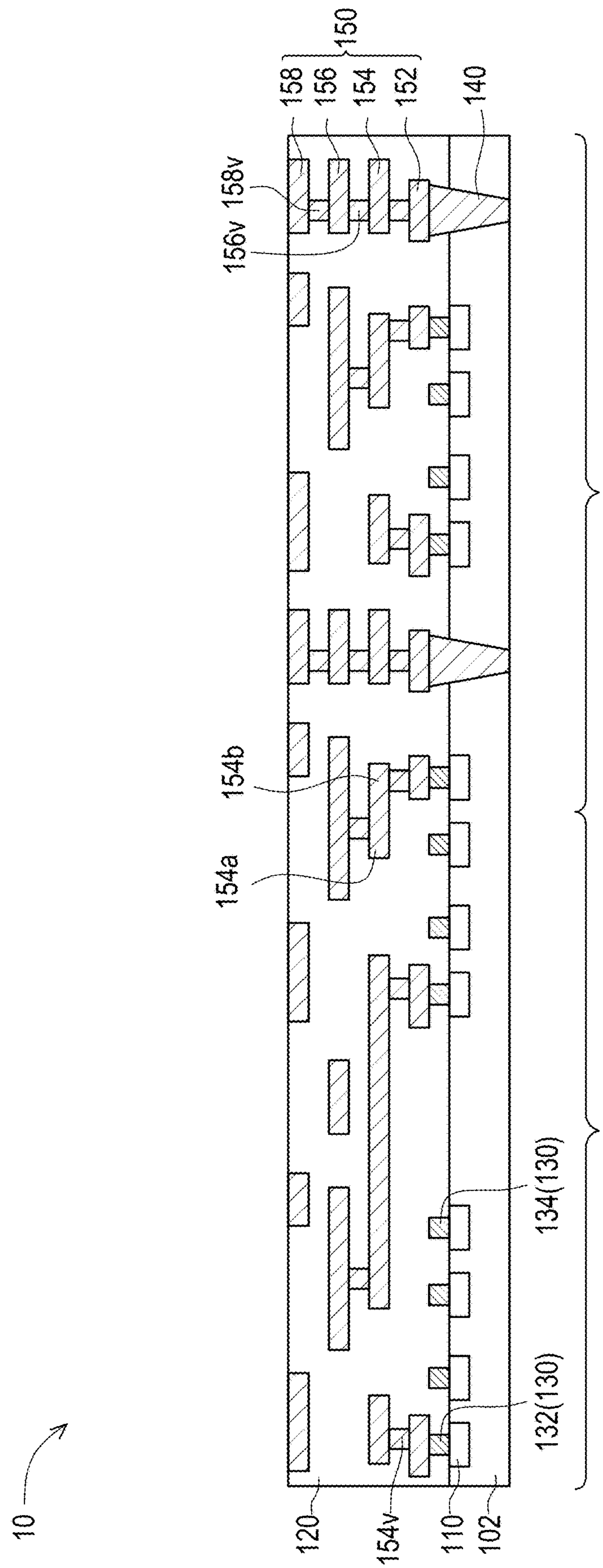

150
158
156
154
152
140
158v
156v
R22
154b
154a
R21
134(130)
132(130)
10
120
154v
110
102

FIG. 12

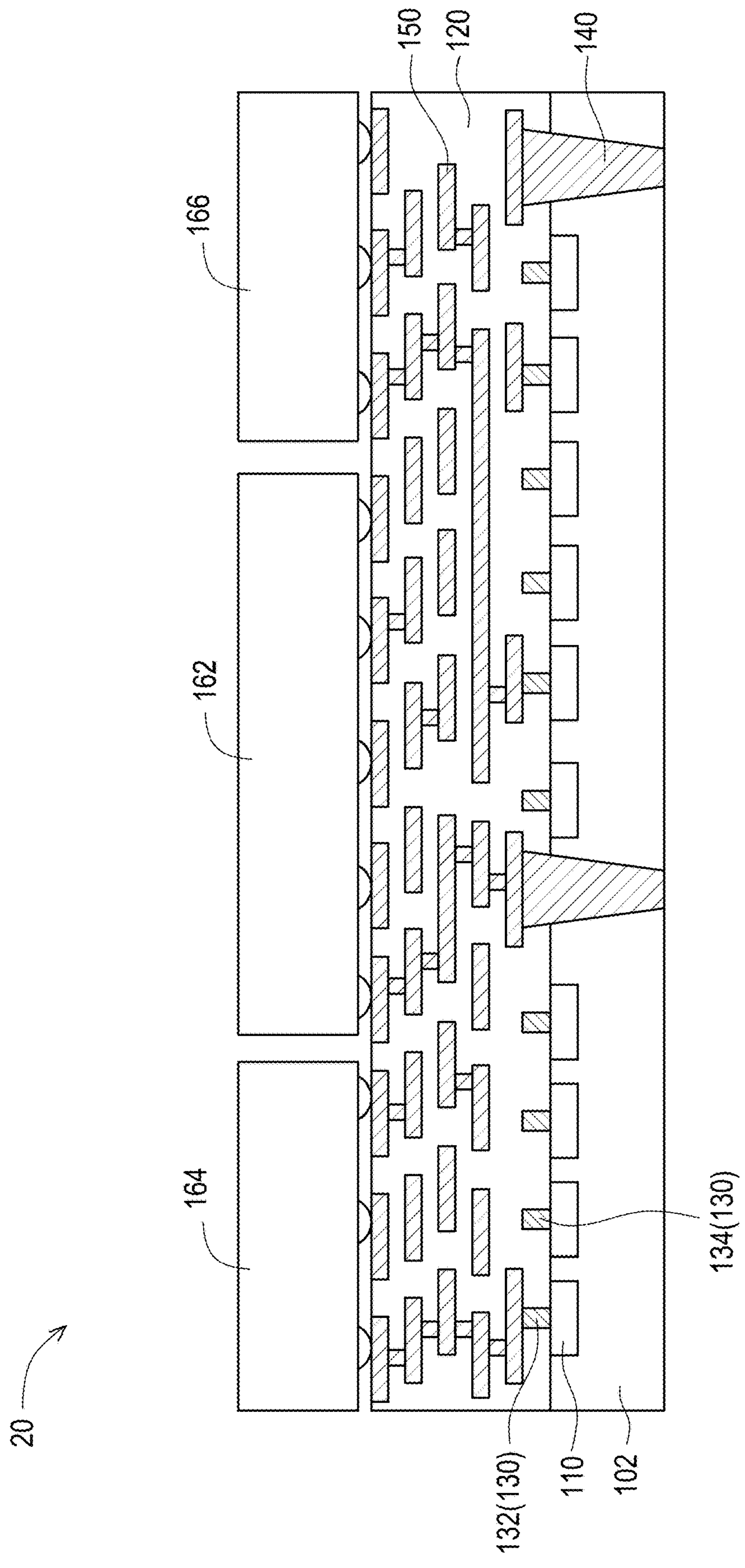


FIG. 13

SEMICONDUCTOR STRUCTURE AND MANUFACTURING METHOD THEREOF

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims the priority benefit of Taiwan application serial no. 113107099, filed on Feb. 27, 2024. The entirety of the above-mentioned patent application is hereby incorporated by reference herein and made a part of this specification.

BACKGROUND

Technical Field

[0002] The disclosure relates to a structure and a manufacturing method thereof, and in particular to a semiconductor structure and a manufacturing method thereof.

Description of Related Art

[0003] With the evolution of technology and market requirements, more and more elements (e.g., memories, logic chips, etc.) need to be packaged together to meet the requirements for high-performance computing. When manufacturing a large die, a single exposure area of a general photomask may not be sufficient to manufacture the large die, and a plurality of photolithography processes by splicing are required to manufacture the large die. In terms of an interposer, as the size increases, the quantity of photomasks required for each layer including a redistribution layer, a via, a passive element, etc. also increases. Further, the photomask for each layer is different for different products, which leads to an increase in manufacturing costs.

SUMMARY

[0004] The disclosure provides a semiconductor structure and a manufacturing method thereof to lower manufacturing costs.

[0005] The manufacturing method of the semiconductor structure of the disclosure includes the following steps. A plurality of passive elements are formed in a substrate. The plurality of passive elements are arranged in an array in the substrate. A plurality of contact members are formed on the plurality of passive elements. The plurality of contact members correspond to the plurality of passive elements. A redistribution layer is formed on the substrate. The redistribution layer is electrically connected to a part of the plurality of passive elements through a part of the plurality of contact members.

[0006] In an embodiment of the disclosure, in a process of forming the plurality of passive elements in the substrate, each different region of the substrate is exposed through a passive element photomask.

[0007] In an embodiment of the disclosure, forming the plurality of contact members on the plurality of passive elements includes respectively exposing different regions of the substrate through a passive element photomask.

[0008] In an embodiment of the disclosure, forming the redistribution layer on the substrate includes enabling a first photomask to correspond to a first subregion, and forming a first feature of the redistribution layer in the first subregion of the substrate through the first photomask; and enabling a second photomask to correspond to a second subregion, and forming a second feature of the redistribution layer in the second subregion of the substrate through the second photomask. A pattern of the first photomask is different from a pattern of the second photomask.

[0009] In an embodiment of the disclosure, the first feature and the second feature are horizontally connected.

[0010] In an embodiment of the disclosure, the substrate includes a plurality of die regions. An area of each of the plurality of die regions is greater than a single maximum exposure area of the first photomask.

[0011] In an embodiment of the disclosure, a size of the passive element photomask is smaller than a size of the first photomask.

[0012] In an embodiment of the disclosure, the part of the plurality of passive elements includes a dummy passive element.

[0013] In an embodiment of the disclosure, the manufacturing method further includes forming a through substrate via in the substrate.

[0014] The semiconductor structure of the disclosure includes a plurality of passive elements, a plurality of contact members, and a redistribution layer. The plurality of passive elements are arranged in an array in a substrate. The plurality of contact members are correspondingly disposed on the plurality of passive elements and in direct contact with the plurality of passive elements. The plurality of contact members include a first contact member and a second contact member. The redistribution layer is disposed on the substrate. The redistribution layer is electrically connected to the corresponding passive element through the first contact member, and the second contact member is not electrically connected to the redistribution layer.

[0015] In an embodiment of the disclosure, the semiconductor structure further includes a dielectric layer disposed on the substrate and laterally surrounding the plurality of contact members and the redistribution layer. A top surface of the second contact member is encapsulated by the dielectric layer.

[0016] In an embodiment of the disclosure, the semiconductor structure further includes a through substrate via disposed in the substrate. A top surface of the plurality of contact members is flush with a top surface of the through substrate via.

[0017] Based on the above, in the disclosure, the plurality of passive elements arranged in the array are formed in the substrate using the same photomask, and the required passive element is selected for an electrical connection using a distribution design of the redistribution layer. As a result, the burden of the layout design of the plurality of passive elements in the substrate can be reduced, and the flexibility of customization can be improved. In terms of manufacturing, the requirement for manufacturing the photomask of the plurality of passive elements can also be reduced, thereby reducing the manufacturing costs.

BRIEF DESCRIPTION OF THE DRAWINGS

[0018] FIGS. **1**A, **1**B, **1**C, **2**A, **2**B, **3**A, **3**B, **4**A, **4**B, **5**A, **5**B, **6**, **7**A, **7**B, **8**A, **8**B, **9**A, **9**B, and **10** to **12** are schematic diagrams of a manufacturing process of a semiconductor structure according to an embodiment of the disclosure.

[0019] FIG. **13** is a cross-sectional schematic diagram of a semiconductor structure according to an embodiment of the disclosure.

DESCRIPTION OF THE EMBODIMENTS

[0020] Reference will now be made to the drawings of the exemplary embodiments of the disclosure to provide a more comprehensive elaboration of the disclosure. However, the disclosure may also be embodied in various forms and is not limited to the embodiments described herein. Thicknesses of the layers and regions in the drawings are exaggerated for clarity. The same or similar reference numerals represent the same or similar elements, which will not be repeated one by one in the following paragraphs.

[0021] It should be understood that, although the terms "first", "second", "third", etc. may be used herein to describe various elements, components, regions, layers, and/or parts, these elements, components, regions, layers, and/or parts should not be limited by these terms. These terms are only used to distinguish one element, component, region, layer, or part from another element, component, region, layer, or part. Thus, the "first element", "component", "region", "layer", or "part" discussed below may be referred to as a second element, component, region, layer, or part without departing from the teachings herein.

[0022] In addition, the directional terms mentioned herein, e.g. "on" or "below", are only used to refer to the directions in the drawings and are not intended to limit the disclosure.

[0023] FIGS. **1**A, **1**B, **1**C, **2**A, **2**B, **3**A, **3**B, **4**A, **4**B, **5**A, **5**B, **6**, **7**A, **7**B, **8**A, **8**B, **9**A, **9**B, and **10** to **12** are schematic diagrams of a manufacturing process of a semiconductor structure according to an embodiment of the disclosure. FIGS. **1**A, **2**A, **3**A, **4**A, **5**A, **7**A, **8**A, **9**A, and **10**A are top-view schematic diagrams. FIGS. **1**B, **2**B, **3**B, **4**B, **5**B, **6**, **7**B, **8**B, **9**B, **11**, and **12** are partial cross-sectional schematic diagrams. FIG. **1**B is a partial cross-sectional schematic diagram of FIG. **1**A. FIG. **1**C is a top-view schematic diagram of a passive element photomask M**1**. FIG. **2**B is a partial cross-sectional schematic diagram of FIG. **2**A. FIG. **3**B is a partial cross-sectional schematic diagram of FIG. **3**A. FIG. **4**B is a partial cross-sectional schematic diagram of FIG. **4**A. FIG. **5**B is a partial cross-sectional schematic diagram of FIG. **5**A. FIG. **7**B is a partial cross-sectional schematic diagram of FIG. **7**A. FIG. **8**B is a partial cross-sectional schematic diagram of FIG. **8**A. FIG. **9**B is a partial cross-sectional schematic diagram of FIG. **9**A. For clarity, some members (e.g., a photoresist layer, a contact member, a dielectric layer, a redistribution layer, etc.) are omitted from FIGS. **1**A, **2**A, **3**A, **4**A, **5**A, **7**A, **8**A, **9**A, and **10**. The omitted parts may be understood by referring to the corresponding cross-sectional diagrams.

[0024] Referring to FIGS. **1**A, **1**B, **1**C, **2**A, **2**B, **3**A, and **3**B, a plurality of passive elements **110** are formed in a substrate **102**. The plurality of passive elements **110** are arranged in an array in the substrate **102**. Specifically, the substrate **102** is provided first. The substrate **102** includes a semiconductor substrate (made of, for example, silicon, silicon germanium, or other suitable semiconductor materials), a silicon-on-insulator (SOI) substrate, or a combination thereof. In some embodiments, a hard mask layer (not shown) may be formed on the substrate **102**, and the disclosure is not limited thereto. In some embodiments, the substrate **102** may be divided into a plurality of die regions **100**, so that the substrate **102** may be cut into a plurality of dies according to the plurality of die regions **100** in a subsequent process. In some embodiments, an area of a single die region **100** is greater than a single maximum exposure area of a general photomask, such as greater than 26 mm*33 mm. In some embodiments, the area of the single die region **100** is about four times the single maximum exposure area of the general photomask. The single maximum exposure area herein refers to a maximum range of the photomask that may be exposed in a single exposure and is independent of a pattern on the photomask.

[0025] Then, different regions of the substrate **102** are respectively exposed through the passive element photomask M**1** using a splicing photolithography process to define positions where the passive elements are subsequently formed. For a pattern layout of the plurality of passive elements **110**, a passive element pattern is basically not arranged in a position that overlaps a predetermined through substrate via. Other than that, the passive element patterns are arranged in an array in all other regions.

[0026] For example, as shown in FIGS. **1**A and **1**B, a first photoresist layer **202** is formed on the substrate **102**. With the passive element photomask M**1** as a mask, the passive element photomask M**1** is enabled to correspond to a first subregion R**11** first to expose the first photoresist layer **202** located in the first subregion R**11**, and an exposure region **202'** is formed in the first photoresist layer **202** in the first subregion R**11**. Then, as shown in FIGS. **2**A and **2**B, the passive element photomask M**1** is enabled to correspond to a second subregion R**12** to expose the first photoresist layer **202** located in the second subregion R**12**, and the exposure region **202'** is formed in the first photoresist layer **202** in the second subregion R**12**. The above steps are repeatedly performed, so that each subregion of the die region **100** is exposed through the passive element photomask M**1**, such that a pattern of the passive element photomask M**1** is transferred to a different position in the die region **100**. That is, each subregion of the die region **100** has the same pattern after being exposed through the passive element photomask M**1**.

[0027] It is noted that in FIG. **1**A, to facilitate description, the single die region **100** is divided into nine subregions (including the first subregion R**11**, the second subregion R**12**, a third subregion R**13**, and a plurality of other unlabeled subregions) with dashed lines. A range of each subregion corresponds to a single maximum exposure range of the passive element photomask M**1**. That is, in FIG. **1**A, the single maximum exposure area of the passive element photomask M**1** is one-ninth of the die region **100**. In other words, the area of the single die region **100** is about nine times the single maximum exposure area of the passive element photomask M**1**. However, the single maximum exposure area of the passive element photomask M**1** (i.e., a size of the passive element photomask M**1**) may be adjusted according to the actual requirements, and the disclosure is not limited thereto. As the single maximum exposure range of the passive element photomask M**1** changes, a quantity of subregions of the single die region **100** varies depending on the single maximum exposure range of the passive element photomask M**1**. In some embodiments, the entire substrate **102**, that is, including the die region **100** and a region outside the die region **100**, is exposed through the passive element photomask M**1**, so that subsequently formed passive elements may substantially cover the entire substrate **102**.

[0028] In some embodiments, as shown in FIG. **1**C, patterns p**1** of the passive element photomask M**1** may be arranged in an array in the passive element photomask M**1**. A partial space of the passive element photomask M**1** may be left without a pattern (e.g., a blank pattern s**1**) to reserve a partial space of the substrate **102** for subsequent use by a through substrate via. In FIG. **1**C, the pattern **p1** and the blank pattern s**1** of the passive element photomask M**1** are only schematically shown and not intended to limit the disclosure. A layout and coverage of the pattern **p1** and the blank pattern s**1** of the passive element photomask M**1** may be adjusted according to the actual requirements.

[0029] Since the area of the single die region **100** is larger than the maximum exposure area achievable in a single exposure by a general photomask that is and each of the plurality of passive elements **110** has the same pattern, the pattern layout of the plurality of passive elements **110** may be obtained through splicing by the same photomask rather than by different photomasks, so that different subregions of the die region **100** are exposed through the same photomask to transfer the pattern layout of the plurality of passive elements **110** to the die region **100**. As a result, a quantity of photomasks required for the passive element **110** can be reduced, thereby reducing manufacturing costs and the number of quality tests of a photomask.

[0030] In some embodiments, the size of the passive element photomask M**1** may be smaller or equal to the size of a general photomask. For example, the size of the passive element photomask M**1** may be 0.1 to 1 times the size of the general photomask.

[0031] Referring to FIGS. **3**A and **3**B, after different regions of the substrate **102** are respectively exposed through the passive element photomask M**1**, the first photoresist layer **202** is developed (e.g., by removing the exposure region **202**' of the first photoresist layer **202**) to obtain a patterned first photoresist layer (not shown). The passive element **110** may be formed in the substrate **102** thereafter by any known method. For example, taking the passive element **110** as a capacitor as an example, the patterned first photoresist layer may be used as a mask to perform etching on the substrate **102** to form a trench (not shown). After the patterned first photoresist layer is removed, a conducting layer, an insulating layer, and a conducting layer (not shown) may be sequentially formed in the trench to form the capacitor embedded in the substrate **102**. However, the disclosure is not limited thereto. Other forming methods of a capacitor are also applicable to this embodiment. In addition, the passive element **110** may be other passive elements such as a resistor and an inductor, and the disclosure is not limited thereto.

[0032] Due to space concerns, eight passive elements **110** are schematically shown in each subregion of the die region **100** in FIG. **3**A. However, it is noted that each subregion of the die region **100** has a passive element **110** corresponding to a position of a pattern of the passive element photomask M**1**, as shown in the partially enlarged view in FIG. **3**A. It is evident that a quantity of the passive element **110** in a single subregion may be adjusted according to the actual requirements by adjusting a pattern layout of the passive element photomask M**1**. In addition, since the passive element photomask M**1** has the blank pattern s**1**, each subregion of the die region **100** also has a region s**2** corresponding to the blank pattern s**1**. The region s**2** may be used as a space for subsequent configuration of a through substrate via. In addition, similar to FIG. **3**A, FIGS. **4**A, **5**A, **7**A, **8**A, **9**A, and **10** in subsequent descriptions also simplify the quantity of elements shown in each subregion in response to space restriction.

[0033] Referring to FIGS. **4**A, **4**B, **5**A, **5**B, and **6**, a plurality of contact members **130** are formed on the plurality of passive elements **110**. The plurality of contact members **130** correspond to the plurality of passive elements **110**. For example, different regions of the substrate **102** are respectively exposed through a contact member photomask M**2** using the splicing photolithography process to define positions where the contact members **130** are subsequently formed. For a pattern layout of the plurality of contact members **130**, a contact member pattern is basically not arranged in a position that overlaps a predetermined through substrate via. The corresponding contact member patterns are arranged in all other regions corresponding to the passive elements **110**.

[0034] Specifically, as shown in FIGS. **4**A and **4**B, a first dielectric layer **122** is formed on the substrate **102**. Then, a second photoresist layer **204** is formed on the first dielectric layer **122**. Thereafter, with the contact member photomask M**2** as a mask, the contact member photomask M**2** is enabled to correspond to the first subregion R**11** first to expose the second photoresist layer **204** located in the first subregion R**11**, and an exposure region **204**' is formed in the first photoresist layer **202** in the first subregion R**11**. Then, as shown in FIGS. **5**A and **5**B, the contact member photomask M**2** is enabled to correspond to a second subregion R**12** to expose the second photoresist layer **204** located in the second subregion R**12**, and the exposure region **204**' is formed in the second photoresist layer **204** in the second subregion R**12**. The above steps are repeatedly performed, so that each subregion of the die region **100** is exposed through the contact member photomask M**2**, such that a pattern of the contact member photomask M**2** is transferred to different positions in the die region **100**. In some embodiments, a single maximum exposure area of the contact member photomask M**2** is basically the same as the single maximum exposure area of the passive element photomask M**1**. Thus, a size of the contact member photomask M**2** is basically the same as the passive element photomask M**1**.

[0035] In some embodiments, the pattern of the contact member photomask M**2** may correspond to the pattern of the passive element photomask M**1** to connect a subsequently formed contact member and the passive element **110**. A partial space of the contact member photomask M**2** may be left without a pattern (e.g., a blank pattern s**3**). The blank pattern s**3** basically corresponds to the blank pattern s**1** of the passive element photomask M**1** or the region s**2** of the die region **100** to reserve a partial space of the substrate **102** for subsequent use by a through substrate via.

[0036] A pattern layout of the plurality of contact members **130** may be obtained by splicing different subregions of the die region **100** through the same photomask, so that different subregions of the die region **100** are exposed through the same photomask, thereby transferring the pattern layout of the plurality of contact members **130** to the die region **100**. As a result, a quantity of photomasks required for the contact member **130** may be reduced, thereby reducing the manufacturing costs and the number of quality tests of a photomask.

[0037] Referring to FIG. **6**, after different regions of the substrate **102** are respectively exposed through the contact member photomask M**2**, the second photoresist layer **204** is developed (e.g., by removing the exposure region **204**' of the second photoresist layer **204**) to obtain a patterned second photoresist layer (not shown). Then, the patterned second photoresist layer may be used as a mask to perform etching on the first dielectric layer **122** to form an opening (not shown). After the patterned first photoresist layer is removed, the opening is filled with a conducting material to form the contact member **130**. The contact member **130** is electrically connected to the corresponding passive element **110**. In FIG. **6**, one contact member **130** is schematically shown to be connected to one passive element **110**. However, this is not intended to limit the disclosure. A quantity of the contact member **130** may be adjusted according to the actual requirements. In addition, since the contact member photomask M**1** has the blank pattern s**3**, each subregion of the die region **100** also has a region corresponding to the blank pattern s**3**. In other words, each subregion of the die region **100** has a reserved space for subsequent configuration of a through substrate via.

[0038] In some embodiments, the corresponding contact member **130** is formed on each of the plurality of passive elements **110**, so that the required passive element **110** may be freely selected to be electrically connected through the corresponding contact member **130** for the subsequent distribution design.

[0039] Referring to FIGS. **7**A and **7**B, a through substrate via **140** is formed in the substrate **102**. For example, the through base via **140** may be included in the second subregion R**12** and the third subregion R**13** of the die region **100**. Specifically, a third photoresist layer **206** may be formed on the first dielectric layer **122** first. Then, a through substrate via photomask M**3** is used as a mask, so that the through substrate via photomask M**3** corresponds to a part of the second subregion R**12** and the third subregion R**13** to expose and develop the third photoresist layer **206** located in the part of the second subregion R**12** and the third subregion R**13**, thereby forming the patterned third photoresist layer **206**. Thereafter, the patterned third photoresist layer **206** may be used as a mask to perform etching on the first dielectric layer **122** and the substrate **102** to form an opening (not shown). Then, the opening is filled with a conducting material to form the through base via **140**. In some embodiments, the patterned third photoresist layer **206** is removed before the opening is filled with the conducting material. Since each subregion of the die region **100** has the reserved space, the through substrate via **140** may be disposed in the spaces according to the actual requirements. It is noted that the through substrate via **140** may be selectively formed in a part, rather than all, of the reserved spaces. In other words, the part of the plurality of reserved spaces may not be provided with the through substrate via **140**, as shown in FIG. **7**A. In addition, the quantity and configuration of the through substrate via **140** formed in the reserved space of each subregion may be adjusted according to the actual requirements, and the disclosure is not limited thereto.

[0040] In some embodiments, a size of the through substrate via photomask M**3** may be greater than the size of the contact member photomask M**2** and the size of the passive element photomask M**1**, but the disclosure is not limited thereto. In other embodiments, the size of the through substrate via photomask M**3** may be equal to the size of the contact member photomask M**2** and the size of the passive element photomask M**1**. In some embodiments, the size of the through substrate via photomask M**3** may be the size of a general photomask. In some embodiments, the area of the single die region **100** is about four times a single maximum exposure area of the through substrate via photomask M**3**.

[0041] FIG. **7**A schematically shows the through substrate via **140** formed in a part of the second subregion R**12** and the third subregion R**13** through the through substrate via photomask M**3**. However, it is noted that when the single maximum exposure area of the single through substrate via photomask M**3** cannot cover a pattern layout of the through substrate via **140**, a plurality of through substrate via photomasks need to be spliced to complete the pattern layout of the through substrate via **140**, and the through substrate via photomasks respectively have different patterns. In this embodiment, the single die region **100** requires splicing of up to four different through substrate via photomasks to complete the pattern layout of the through substrate via.

[0042] Referring to FIGS. **8**A, **8**B, **9**A, **9**B, **10**, and **11**, a redistribution layer **152** is formed on the substrate **102**. The redistribution layer **152** is electrically connected to a part of the passive element **110** through a part of the contact member **130**. Since the area of the single die region **100** is greater than the single maximum exposure area of the general photomask, a pattern layout of the redistribution layer **152** requires a plurality of exposures through a plurality of different redistribution circuit photomasks to transfer the pattern layout of the redistribution layer **152** to the die region **100**.

[0043] Specifically, as shown in FIGS. **8**A and **8**B, a second dielectric layer **124** is formed on the first dielectric layer **122**. Then, a fourth photoresist layer **208** is formed on the second dielectric layer **124**. Thereafter, with a first photomask M**4***a* as a mask, the first photomask M**4***a* is enabled to correspond to a first subregion R**21** first to expose the fourth photoresist layer **208** located in the first subregion R**21**, and an exposure region **208'** is formed in the fourth photoresist layer **208** in the first subregion R**21**. Then, as shown in FIGS. **9**A and **9**B, a second photomask M**4***b* is enabled to correspond to a second subregion R**22** to expose the fourth photoresist layer **208** located in the second subregion R**22**, and the exposure region **208'** is formed in the fourth photoresist layer **208** in the second subregion R**22**. Similarly, as shown in FIG. **10**, a third subregion R**23** and a fourth subregion R**24** are sequentially exposed through a third photomask M**4***c* and a fourth photomask M**4***d*, respectively. Thereafter, referring to FIG. **11**, the fourth photoresist layer **208** is developed (e.g., by removing the exposure region **208'** of the fourth photoresist layer **208**) to form a patterned fourth photoresist layer (not shown). With the patterned fourth photoresist layer as a mask, etching is performed on the second dielectric layer **124** to form an opening (not shown). After the patterned fourth photoresist layer is removed, the opening is filled with a conducting material to form the redistribution layer **152**.

[0044] The first photomask M**4***a*, the second photomask M**4***b*, the third photomask M**4***c*, and the fourth photomask M**4***d* are different photomasks of the same size, each of which has a different pattern. The patterns of the first photomask M**4***a*, the second photomask M**4***b*, the third photomask M**4***c*, and the fourth photomask M**4***d* are combined to constitute the pattern layout of the redistribution layer **152**. As a result, the redistribution layer **152** may be formed in a feature size that is greater than a single maximum achievable exposure range of a general photomask in length or area.

[0045] It is noted that in FIGS. **8**A, **9**A, and **10**, to facilitate description, the single die region **100** is divided into four subregions (including the first subregion R**21**, the second subregion R22, the third subregion R23, and the fourth subregion R24) with dashed lines. A range of each subregion corresponds to a single maximum exposure range of the first photomask M4*a* (or the second subregion R22, the third subregion R23, or the fourth subregion R24). That is, in FIG. 8A, the single maximum exposure area of the first photomask M4*a* is one-fourth of the die region **100**. In other words, the area of the single die region **100** is about four times the single maximum exposure area of the first photomask M4*a*. However, the single maximum exposure range of the first photomask M4*a* (i.e., a size of the first photomask M4*a*) may be adjusted according to the actual requirements, and the disclosure is not limited thereto. As the single maximum exposure range of the first photomask M4*a* changes, a quantity of subregions of the single die region **100** varies depending on the single maximum exposure range of the first photomask M4*a*.

[0046] In some embodiments, the size of the passive element photomask M1 is smaller than the size of the first photomask M4*a*, a size of the second photomask M4*b*, a size of the third photomask M4*c*, and a size of the fourth photomask M4*d*. For example, the size of the passive element photomask M1 may be 0.1 to 1 times the size of the first photomask M4*a*. In other embodiments, the size of the passive element photomask M1 is equal to the size of the first photomask M4*a*, the size of the second photomask M4*b*, the size of the third photomask M4*c*, and the size of the fourth photomask M4*d*.

[0047] In some embodiments, a part of the redistribution layer **152** is electrically connected to a part of the passive element **110** through a part of the contact member **130**. Since the substrate **102** includes the plurality of passive elements **110** and the corresponding contact members **130**, the redistribution layer **152** may be electrically connected to a part of the passive element **110** according to the distribution requirements, thereby improving the convenience of customized design. Also, the number of times of layout redesigns and quality tests of the passive element **110** and the contactor member **130** for each new product development can be reduced.

[0048] In some embodiments, the passive element **110** that is not electrically connected to the redistribution layer **152** and the corresponding contact member **130** are dummy elements.

[0049] Referring to FIG. **12**, a plurality of redistribution layers **154**, **156**, and **158** are formed on the redistribution layer **152**. For example, a dielectric layer (not individually labeled) may be formed on the redistribution layer **152** first. Then, an opening (not shown) is formed in the dielectric layer through a conductive via photomask (not shown). Thereafter, a conductive material layer (not shown) is formed on the dielectric layer and in the opening, and the conductive material layer is patterned through a redistribution circuit photomask (not shown) to form the redistribution layer **154** and a conductive via **154***v*. Similar steps as described above are repeated to successively form the redistribution layers **156** and **158**, a conductive via **156***v* connected between the redistribution layer **154** and the redistribution layer **156**, and a conductive via **158***v* connected between the redistribution layer **156** and the redistribution layer **158** located in the dielectric layer **120**. It is noted that the dielectric layer **120** in FIG. **12** is formed by stacking a plurality of dielectric layers (including the first dielectric layer **122** and the second dielectric layer **124**). The plurality of redistribution layers **152**, **154**, **156**, and **158** are collectively referred to as a redistribution layer **150**.

[0050] Since the area of the single die region **100** is greater than the maximum achievable exposure area in a single exposure of a general photomask, the pattern layout of each of the conductive vias **154***v*, **156***v*, and **158***v* needs to be completed through splicing four different conductive via photomasks. The pattern layout of each of the redistribution layers **152**, **154**, **156**, and **158** needs to be completed through splicing four different redistribution circuit photomasks.

[0051] In some embodiments, a first feature **154***a* of the redistribution layer **154** located in the first subregion R21 may be formed through a first redistribution circuit photomask (not shown), while a second feature **154***b* of the redistribution layer **154** located in the second subregion R22 may be formed through a second redistribution circuit photomask (not shown). The first feature **154***a* and the second feature **154***b* are horizontally connected. As can be seen, the redistribution layer **154** may be formed in a feature size surpassing the limitation of a single maximum exposure range of a general photomask through a photomask splicing manner.

[0052] Four redistribution layers are schematically shown in FIG. **12**, but not intended to limit the disclosure. The quantity of redistribution layers and the distribution design may be adjusted according to the actual requirements.

[0053] In some embodiments, the quantity and locations of through substrate via patterns in the die region **100** may be determined first during the design stage of a layout of the die region **100**. Then, a plurality of passive element patterns may be arranged in an array in the remaining substrate **102** in the die region **100**, and a space may be reserved according to the pattern layout of the through substrate via. On the other hand, the distribution design of each layer of the redistribution layer **150** may be performed based on the linkage requirements. Thereafter, the redistribution layer **150** is electrically connected to the passive element **110** at a proper location according to the distribution design of the redistribution layer **150** at the bottommost layer (e.g., the redistribution layer **152**). As a result, during new product development, the distribution design of the redistribution layer may be adjusted without redesigning the layout of the passive element **110** and the contact member **130**, thereby easing the burden on designers and reducing development costs.

[0054] In some embodiments, a planarization process may be performed on a bottom surface of the substrate **102** to expose a bottom surface of the through substrate via **102**. Then, a conductive connecting member (not shown) may be formed on the bottom surface of the through substrate via **102** to be electrically connected to the outside.

[0055] In some embodiments, a singulation process may be performed to partition the substrate **102** and a structure thereon into a plurality of semiconductor structures **10** according to the die region **100**.

[0056] Through the above processes, the manufacturing method of the semiconductor structure **10** is generally completed.

[0057] The semiconductor structure **10** includes the plurality of passive elements **110**, the plurality of contact members **130**, and the redistribution layer **150** (including the redistribution layer **152**, the redistribution layer **154**, the redistribution layer **156**, and the redistribution layer **158**). The plurality of passive elements **110** are arranged in an array in the substrate **102**. The plurality of contact members **130** are correspondingly disposed on the plurality of passive elements **110** and in direct contact with the plurality of passive elements **110**. The plurality of contact members **130** include a first contact member **132** and a second contact member **134**. The redistribution layer **150** is disposed on the substrate **102**. The redistribution layer **150** is electrically connected to the corresponding passive element **110** through the first contact member **132**, and the second contact member **134** is not electrically connected to the redistribution layer **150**.

[0058] In some embodiments, the plurality of conductive vias **154*v***, **156*v***, and **158*v*** are connected between the adjacent redistribution layers **150** to provide the redistribution layer **150** with an electrical connection in a vertical direction.

[0059] In some embodiments, the passive element **110** in direct contact with the second contact member **134** is a dummy passive element.

[0060] In some embodiments, the semiconductor structure **10** further includes the dielectric layer **120**. The dielectric layer **120** is disposed on the substrate **102** and laterally surrounds the plurality of contact members **130** and the redistribution layer **150**. A top surface of the second contact member **134** is encapsulated by the dielectric layer **120**. That is, the second contact member **134** is a dummy contact member.

[0061] In some embodiments, the semiconductor structure **10** further includes the through substrate via **140** disposed in the substrate **102** to provide electrical connections on both sides of the substrate **102**.

[0062] In some embodiments, the through substrate via **140** is also embedded in a part of the dielectric layer **120**. The top surface of the contact member **130** may be flush with the top surface of the through substrate via **140**.

[0063] In some embodiments, the semiconductor structure **10** may serve as an interposer to be connected between a chip and a circuit board.

[0064] FIG. **13** is a cross-sectional schematic diagram of a semiconductor structure according to an embodiment of the disclosure. It should be noted herein that, the embodiment of FIG. **13** continues to use the reference numerals and some content of the embodiment of FIG. **12**, wherein the same or similar reference numerals are adopted to represent the same or similar elements, and description of the same technical content is omitted. For the description of the omitted part, reference may be made to the foregoing embodiment, which will not be repeated.

[0065] Referring to FIG. **13**, a semiconductor structure **20** includes the plurality of passive elements **110**, the plurality of contact members **130**, and the redistribution layer **150**. The plurality of passive elements **110** are arranged in an array in the substrate **102**. The plurality of contact members **130** are correspondingly disposed on the plurality of passive elements **110** and in direct contact with the plurality of passive elements **110**. The plurality of contact members **130** include a first contact member **132** and a second contact member **134**. The redistribution layer **150** is disposed on the substrate **102**. The redistribution layer **150** is electrically connected to the corresponding passive element **110** through the first contact member **132**, and the second contact member **134** is not electrically connected to the redistribution layer **150**.

[0066] The semiconductor structure **20** is generally similar to the semiconductor structure **10**. The semiconductor structure **20** differs from the semiconductor structure **10** in that the semiconductor structure **20** further includes a plurality of chips **162**, **164**, and **166** disposed on the dielectric layer **120**. The plurality of chips **162**, **164**, and **166** may be electrically connected to the redistribution layer **150** though a conductive connecting member (not labeled).

[0067] In some embodiments, the chip **162** is located between the chip **164** and the chip **166**. In some embodiments, the chip **162** may be a system on chip, and the chips **164** and **166** may be high bandwidth memories. However, the disclosure is not limited thereto.

[0068] In summary, in the disclosure, the plurality of passive elements arranged in the array are formed in the substrate using the same photomask, and the required passive element is selected for an electrical connection using the distribution design of the redistribution layer. As a result, the burden of the layout design of the plurality of passive elements in the substrate can be reduced, and the flexibility of customization can be improved. In terms of manufacturing, the requirement for manufacturing the photomask of the plurality of passive elements can also be reduced, thereby reducing the manufacturing costs.

[0069] Although the disclosure has been disclosed in the above embodiments, the embodiments are not intended to limit the disclosure. Persons skilled in the art may make some changes and modifications without departing from the spirit and scope of the disclosure. Therefore, the protection scope of the disclosure shall be defined by the appended claims.

What is claimed is:

1. A manufacturing method of a semiconductor structure, comprising:

forming a plurality of passive elements in a substrate, wherein the plurality of passive elements are arranged in an array in the substrate;

forming a plurality of contact members on the plurality of passive elements, the plurality of contact members corresponding to the plurality of passive elements; and

forming a redistribution layer on the substrate, wherein the redistribution layer is electrically connected to a part of the plurality of passive elements through a part of the plurality of contact members.

2. The manufacturing method of the semiconductor structure of claim **1**, wherein forming the plurality of passive elements in the substrate comprises:

respectively exposing different regions of the substrate through a passive element photomask.

3. The manufacturing method of the semiconductor structure of claim **1**, wherein forming the plurality of contact members on the plurality of passive elements comprises:

respectively exposing different regions of the substrate through a contact member photomask.

4. The manufacturing method of the semiconductor structure of claim **2**, wherein forming the redistribution layer on the substrate comprises:

enabling a first photomask to correspond to a first subregion, and forming a first feature of the redistribution layer in the first subregion of the substrate through the first photomask; and

enabling a second photomask to correspond to a second subregion, and forming a second feature of the redistribution layer in the second subregion of the substrate through the second photomask, wherein a pattern of the first photomask is different from a pattern of the second photomask.

5. The manufacturing method of the semiconductor structure of claim **4**, wherein the first feature and the second feature are horizontally connected.

6. The manufacturing method of the semiconductor structure of claim **4**, wherein the substrate comprises a plurality of die regions, and an area of each of the plurality of die regions is greater than a single maximum exposure area of the first photomask.

7. The manufacturing method of the semiconductor structure of claim **6**, wherein a size of the passive element is smaller than a size of the first photomask.

8. The manufacturing method of the semiconductor structure of claim **1**, wherein a part of the plurality of passive elements is a dummy passive element.

9. The manufacturing method of the semiconductor structure of claim **1**, further comprising: forming a through substrate via in the substrate.

10. A semiconductor structure, comprising:

a plurality of passive elements, arranged in an array in a substrate;

a plurality of contact members, correspondingly disposed on the plurality of passive elements and in direct contact with the plurality of passive elements, wherein the plurality of contact members comprise a first contact member and a second contact member; and

a redistribution layer, disposed on the substrate, wherein the redistribution layer is electrically connected to the corresponding passive element through the first contact member, and the second contact member is not electrically connected to the redistribution layer.

11. The semiconductor structure of claim **10**, further comprising:

a dielectric layer, disposed on the substrate and laterally surrounding the plurality of contact members and the redistribution layer, wherein a top surface of the second contact member is encapsulated by the dielectric layer.

12. The semiconductor structure of claim **10**, further comprising:

a through substrate via, disposed in the substrate, wherein a top surface of the plurality of contact members is flush with a top surface of the through substrate via.

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